Two kinds of rules, in plain English

Rule languages · Logical English · Logic Production Systems

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An Act of Parliament

> **British Nationality Act 1981, section 1**
>
> (1) A person born in the United Kingdom after commencement shall be a British citizen if at the time of the birth his father or mother is—
>
> (a) a British citizen; or
>
> (b) settled in the United Kingdom.

In 1986 we wrote this Act as a logic program, in Prolog. Forty years later the program is English again.



1 · Rule languages

Two kinds of rules



The Act has two kinds of sentence

WHAT IS THE CASE

> **1(1)** A person born in the United Kingdom … shall be a British citizen if …

It describes the world.

It is true or it is false.

It is a **belief**.

WHAT TO DO

> **12(1)** If any British citizen … makes … a declaration of renunciation …, the Secretary of State shall cause the declaration to be registered.

It does not describe the world.

It asks for the world to be changed.

It is a **goal**.

Belief rules

conclusion **if** *conditions*

```
a person is a British citizen
    if the person is born in the UK
    and a parent of the person is a British citizen.
```

- A belief rule defines its conclusion in terms of its conditions
- A set of belief rules is a **logic program**
- The same rule is used **backwards**, to answer a question, and **forwards**, to derive a consequence



Goal rules

if *conditions* **then** *conclusion*

```
if a British citizen declares renunciation
then the Secretary of State registers the declaration.
```

```
it must not be true that
    the Secretary of State registers the declaration of a person
    and it is not the case that the person has another nationality.
```

- A goal rule does not define anything. It must be **made true**, by acting
- The first kind is a **reactive rule**. The second kind is a **constraint**
- An obligation is a goal. A prohibition is a constraint



One section, three kinds of sentence

> **12.**—(1) If any British citizen of full age and capacity makes in the prescribed manner a declaration of renunciation of British citizenship, then, subject to subsections (3) and (4), the Secretary of State shall cause the declaration to be registered.
>
> (2) On the registration of a declaration made in pursuance of this section the person who made it shall cease to be a British citizen.
>
> (3) A declaration made by a person in pursuance of this section shall not be registered unless the Secretary of State is satisfied that the person who made it will after the registration have or acquire some citizenship or nationality other than British citizenship;
>
> …

(1) is a goal: an obligation. **(2)** is a belief: what an event causes. **(3)** is a goal: a constraint.

Rule languages take sides

Mostly belief rules	Mostly goal rules
Prolog, Datalog, answer set programming, s(CASP)	Production rules: Drools
Catala, OpenFisca, RuleSpec	Event-condition-action rules
Decision tables	Obligations and prohibitions: LegalRuleML, L4
Blawx, Epilog, Oracle Intelligent Advisor	Smart contracts: Solidity, Daml

A language with one kind of rule must imitate the other kind.

Logical English writes belief rules. **Logic Production Systems** adds goal rules, and time.



2 · Controlled natural language and Logical English

English that a computer can read as logic

English is expressive. It is also ambiguous.

> A person born in the United Kingdom after commencement shall be a British citizen if at the time of the birth his father or mother is (a) a British citizen; or (b) settled in the United Kingdom.

- **His** father: whose?
- **The** birth: which birth?
- A citizen **when**?

A CONTROLLED NATURAL LANGUAGE

A part of English with one reading only.

LOGICAL ENGLISH

- *conclusion* **if** *conditions*
- No pronouns: **a** person, then **the** person
- One word, one meaning
- Time, but no tense

It is another way to write a logic program.

Templates

```
the templates are:
    *a person* is a British citizen.
    *a person* is born in *a place* on *a date*.
    *a date* is after commencement.
    *a person* is a parent of *a child*.
    *a person* is a British citizen on *a date*.
    *a person* is settled in the UK on *a date*.
```

- A **template** is the pattern of a sentence
- The words between asterisks are **places**. Each place stands for one thing
- In logic, a template is a predicate, and a place is an argument



Section 1(1), in Logical English

```
rule section_1_1 with provenance the British Nationality Act 1981 at "section 1(1)":
a person is a British citizen
    if the person is born in the UK on a date
    and the date is after commencement
    and a parent is a parent of the person
    and the parent is a British citizen on the date
        or the parent is settled in the UK on the date.
```

- **a** person introduces a variable. **the** person refers back to it
- "his father or mother" is now **a parent** of **the person**
- "at the time of the birth" is now **the date**, said each time
- The first line is a **citation**: an explanation shows where the rule comes from



A scenario is a case. A query is a question.

```
scenario john is:
    john is born in the UK on 1990-05-01.
    mary is a parent of john.
    mary is a British citizen on 1990-05-01.
    citizens expects answers ["john is a British citizen"].

query citizens is:
    which person is a British citizen.
```

- A **scenario** gives the facts of one case
- A scenario can say which answers it **expects**. Then the scenario is also a test



Every answer has an explanation

WHY IS JOHN A BRITISH CITIZEN?

```
john is a British citizen
  john is born in the UK on 1990-05-01
  1990-05-01 is after commencement
    1990-05-01 is after 1982-12-31
  mary is a parent of john
  mary is a British citizen on 1990-05-01
```

WHY IS PAUL NOT?

```
scenario paul is:
    paul is born in France on 1990-05-01.
    mary is a parent of paul. …
```

```
not: a person is a British citizen
  not: a person is born in the UK on a date
```

The explanation is the proof. It uses the sentences of the program, so the reader of the rules can read it.



A conditional answer

```
scenario peter is:
    peter is born in the UK on 1990-05-01.
    ann is a parent of peter.
    it is unknown whether ann is settled in the UK on 1990-05-01.
```

Answer: peter is a British citizen, **assuming** ann is settled in the UK on 1990-05-01.

- The case does not say whether ann is settled. The program does not pretend to know
- The answer names what it depends on. A person, or a court, decides the rest
- In logic, this is **abduction**: reasoning with assumptions



3 · Logic Production Systems

Rules in a world that changes



Citizenship changes

```
the events are:
    *a person* is born in the UK
        to *a parent*.
    *a person* declares renunciation.
    *a person* acquires
        another nationality.

the actions are:
    the Secretary of State registers
        the declaration of *a person*.

the fluents are:
    *a person* is a British citizen.
    *a person* is settled in the UK.
    *a person* has another nationality.
```

- A **fluent** is a fact that changes over time
- An **event** happens, and changes the fluents
- An **action** is an event that an agent chooses

The same language. One new first line:

```
the target language is: lps.
```



Causal laws: what an event does

```
when a person is born in the UK to a parent
    and the parent is a British citizen
then the person is a British citizen.

when a person is born in the UK to a parent
    and the parent is settled in the UK
then the person is a British citizen.

when the Secretary of State registers the declaration of a person
then it is not the case that the person is a British citizen.
```

- A **causal law** says which fluents an event starts, and which fluents it ends
- Section 1(1) is now two causal laws. Section 12(2) is the third



Goals: a reactive rule and a constraint

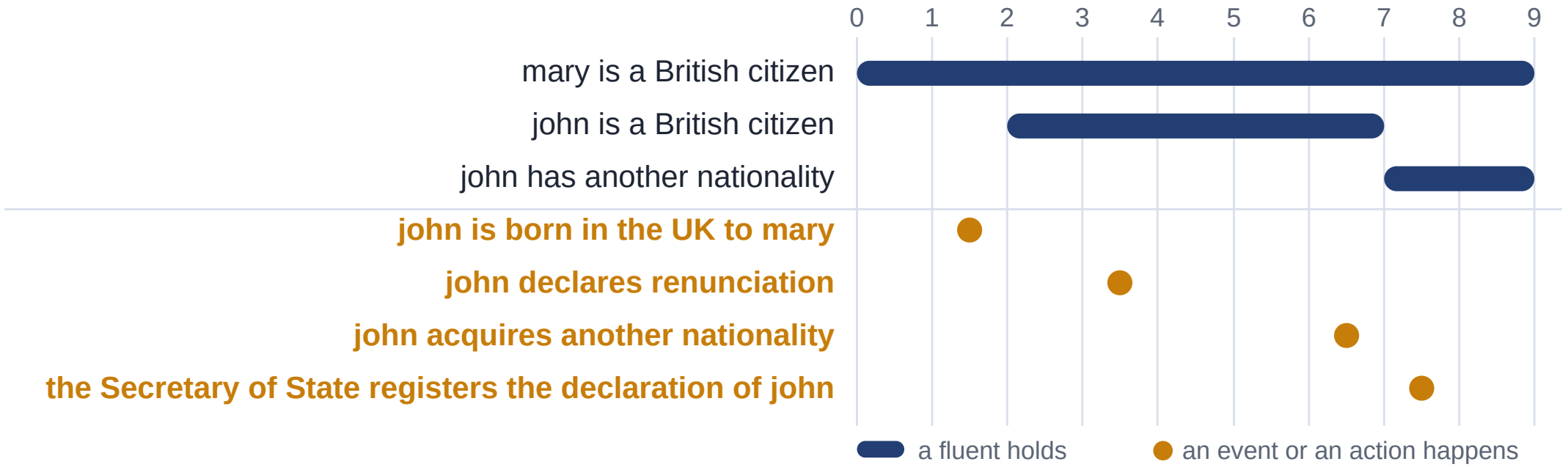
```
if a person is a British citizen at a first time
    and the person declares renunciation from the first time to a second time
then the Secretary of State registers the declaration of the person
        from a third time to a fourth time.

it must not be true that
    the Secretary of State registers the declaration of a person
    and it is not the case that the person has another nationality.
```

- The reactive rule is section 12(1): the Secretary of State **shall** register
- The constraint is section 12(3): **shall not** be registered **unless** …
- The obligation has no deadline. The constraint says when the action may happen



The life of a citizen



The events come from outside. The rules choose the actions. The timeline is the model that makes the goals true.

What to take away

- **Belief rules** say what is the case: *conclusion* if *conditions*
- **Goal rules** say what must be made true: if *conditions* then *conclusion*
- **Logical English** writes rules in English that has one reading only
- **Logic Production Systems** makes the goals true in a world that changes

Next: sixteen other rule languages, read in the same English.

